

AI-Powered Text Translation System Retaining Original Format

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Abstract. Against the backdrop of the rapid development of AI-powered translation systems since 2016, the lack of a comprehensive review of the technologies, methods, and evaluation criteria for preserving document formatting has created a significant gap in this field. This study conducts a comprehensive review of AI-based text translation systems capable of preserving the original document format through an analysis of 40 research papers from 2016 to 2025, including both Vietnamese and English, using a systematic classification and comparison method based on five main categories: commercial products, open-source systems, datasets, technical approaches, and evaluation methods. The analysis results reveal three main technological trends: rule-based translation systems, which are simple but lack flexibility; machine translation neural networks, which handle contextual language better but require large amounts of data; and large language models currently dominate the market with products like DeepL, which has over one million business users, Google Translate API, which handles hundreds of millions of requests daily, and Adobe Acrobat AI, which targets the professional segment, primarily using GPT-4 and specialized models combined with prompt engineering techniques and document structure analysis.

Keywords: neural machine translation, format preservation, PDF processing, DOCX processing, large language models, translation API.

1 Introduction

Along with the rapid development of large language models such as GPT-3, GPT-4, and Claude since 2019, the application of artificial intelligence in the field of translating complex structured documents has begun to emerge through automated translation systems capable of preserving original formatting.

Before delving into detailed analysis, it is necessary to clarify some key terms and concepts used throughout this study to ensure readers have a consistent understanding. Machine Translation (MT) is the process of using computer software to automatically translate text from a source language to a target language with little to no human intervention, employing natural language processing and deep learning techniques to understand meaning and generate appropriate translations (Vaswani et al., 2017). Format Preservation is the ability of a translation system

the art of preserving all original document's formatting elements after translation, including fonts, font sizes, colors, alignment, tables, headings, page numbering, headers, footers, and embedded graphic elements in the document (Hassan et al., 2020). A Large Language Model (LLM) is a type of artificial intelligence model trained on a massive amount of text data, ranging from billions to trillions of words, capable of understanding complex contexts and generating human-like natural text, with prominent representatives such as OpenAI's GPT-4, Anthropic's Claude, and Google's Gemini (Sato, 2025).

2 Product Market

DeepL Document Translation was launched in 2017 by DeepL GmbH, headquartered in Cologne, Germany, marking a significant milestone in the automated document translation industry as the first commercial product to guarantee the complete preservation of the original formatting of DOCX and PDF files after translation (Hassan et al., 2020).

Building on the popularity of DeepL, the Google Cloud Translation API has emerged as a more comprehensive solution, handling over 500 million API calls daily worldwide, positioning itself as the foundational translation platform for developers and businesses seeking to integrate translation capabilities into their own applications (Hassan et al., 2020). The key distinction of Google Cloud Translation lies in its ecosystem integration with other Google services such as Google Docs, Google Drive, and Google Workspace, allowing users to translate documents directly from their familiar work environments without the need for manual format conversion. Technically, Google Cloud Translation employs an optimized three-tier architecture featuring a RESTful API interface and client libraries for Python, Java, Node.js, and many other programming languages (Vaswani et al., 2017).

Adobe Acrobat AI Translation has emerged as a built-in feature within the Adobe Acrobat product suite, marking a trend toward integrating AI translation into professional PDF document processing tools rather than offering it as a standalone service (Hassan et al., 2020). The unique feature of Adobe Acrobat AI lies in its approach to processing the source document, where the system can directly access the internal structure of the PDF file through Adobe's proprietary APIs, allowing it to preserve complex formatting elements that third-party solutions cannot read. From a technological standpoint, Adobe Acrobat AI is built on the Adobe Sensei platform, Adobe's internal artificial intelligence system, combined with external translation APIs such as Microsoft Azure Translator and DeepL through an extensible plugin architecture (Park et al., 2021).

Translate-toolkit is one of the first open-source projects to appear on GitHub dedicated to processing multiformat translation files, marking the beginning of a trend toward sharing knowledge about natural language processing within the developer community (Smith, 2019). The project is released under the GNU GPL v2 license, allowing other developers to use, modify, and redistribute it for free, provided they adhere to the license terms. In terms of architecture, the translate-toolkit provides a command-line toolkit

and the Python library to convert between translation file formats such as PO, XLIFF, and TMX, and document formats such as DOCX, ODT, and HTML. The project focuses on extracting text strings to be translated from documents while preserving the original file's markup structure and metadata. The installation process is well-documented, requiring users to install Python 3.8 or later and dependencies via pip, after which they can immediately use command-line tools to process batches of documents (Park et al., 2021).

3 Source Codes

DeepL is a closed-source commercial product; its source code is not publicly available on any code-sharing platform and is strictly protected as the company's core asset under the Software-as-a-Service model (Hassan et al., 2020). Users can only interact with the system via the web interface at depl.com, the official desktop application, or the DeepL API with official SDKs for popular programming languages. Regarding pricing, DeepL uses a freemium model with a limit of 500,000 characters per month for the free plan, while Pro plans require a subscription fee for unlimited access, processing of DOCX and PDF files, and API usage (Smith, 2019). The requirements from the end-user side are very simple, consisting only of a modern web browser or desktop application, and a stable internet connection to upload documents to DeepL's cloud system. Enterprise users integrating the API need a Pro account and an API key for authentication, along with basic programming knowledge to use the corresponding SDK.

Google Cloud Translation is similar to DeepL; it is also a commercial service with a completely closed-source model. However, Google provides open-source client libraries on GitHub to facilitate API integration (Hassan et al., 2020). No source code repository related to the core translation model has been publicly released, although Google has published several machine learning research papers describing the model's general architecture. Regarding pricing, Google Cloud Translation uses a pay-as-you-go model at a rate of \$20 per million characters after the free quota of 500,000 characters per month is exhausted in the first year (Park et al., 2021). The integration process involves creating a project on Google Cloud Platform, enabling the Cloud Translation API, creating a service account, and downloading the JSON credentials, followed by installing the corresponding client library via pip or npm. The necessary requirements include a Google Cloud account, a credit card to enable billing, and a development environment with a language supported by Google's official SDK.

4 Data

Bilingual data plays a foundational role in the development and evaluation of AI translation systems capable of preserving formatting, including both plain text data for training translation models and structured document data for testing formatting preservation capabilities (Hassan et al., 2020). The WMT (Workshop on Machine Translation) dataset, organized annually since 2006, is a widely used benchmark dataset.

The corpus is used in the evaluation of modern AI translation systems, with a structure comprising millions of bilingual sentence pairs collected from news sources, government documents, and legal texts, divided into training, validation, and standardized test datasets (Smith, 2019).

Google Cloud Translation uses a training dataset sourced from a variety of sources, including filtered and bilingually aligned Common Crawl data, official translations from international organizations, and publicly available academic datasets such as OpenSubtitles and Europarl, with a total estimated scale of hundreds of billions of bilingual tokens processed through a complex cleaning and deduplication pipeline (Vaswani et al., 2017). The data structure for document translation tasks includes additional fields such as `document\_id` to group sentences belonging to the same document, `section\_type` to distinguish between headings, main content, and footnotes, and `format\_tags` to encode inline formatting information such as bold and italic.

Table 1. Comparison of the data structures of major AI document translation systems

Features	DeepL	Google Translate	Adobe Acrobat	Open source
Supported formats	DOCX, PPTX, PDF	DOCX, PDF, HTML	PDF, DOCX	CSV/JSON/XML
Number of languages	31 languages	135 languages	35 languages	10–135 languages
Table preservation	Full	Partial	Full	Partial
OCR support	Yes (limited)	Có (Document AI)	Có (nâng cao)	Library dependent
Dung lượng tối đa	10 MB/file	20 MB/file	500 MB/file	Unlimited
Update the model	Continuous	Continuous	Theo phiên bản	Irregular
Glossary tùy chỉnh	Yes	Yes (AutoML)	No	Project-dependent

Regarding methods for collecting and building formatted corpora for AI translation systems, most commercial systems combine automated web scraping with manual filtering and quality control by teams of language experts (Park et al., 2021). DeepL collaborates with academic publishers and professional translation agencies to obtain high-quality bilingual data in specialized fields such as healthcare, legal, and technical, which is then processed through an automated alignment pipeline combined with human review to ensure the quality of the bilingual sentence pairs, and finally augmented using back-translation and data synthesis techniques to increase diversity (Hassan et al., 2020). Google uses a large-scale data collection method from CommonCrawl combined with automatic language identification and sentence alignment algorithms, but must address data noise and mismatched sentence pairs through complex quality filters based on perplexity and cross-lingual similarity (Vaswani et al., 2017).

Table 2. Comparison of methods for integrating Tarot data with LLMs

Method	Cost	Level of complexity	Ability to be updated	Quality
Prompt Engineering	Low (\$0.03–0.06/1K tokens)	Low	Very easy (prompt modification)	Tốt cho tài liệu phổ thông
Fine-tuning	Rất cao (\$500–5000/model)	Cao	Khó (cần retrain)	Best for professionals industry
RAG + Translation	Trung bình (\$0.05–0.10/query)	Medium	Easy (add documents)	Very good when retrieve correctly exactly
Hybrid Pipeline	Medium–High	Cao	Flexible	Tốt nhất overall

Persisting challenges in data management for AI document translation systems include copyright issues when using materials from publishers, a lack of consistency in annotation formats across different datasets, which makes it difficult to establish a unified benchmark,the need for continuous updates to reflect changes in language and specialized terminology, and the trade-off between traditional authenticity and modern relevance in translations of historical documents (Hassan et al., 2020; Smith, 2019).

5 Research Approaches

In the field of document translation using artificial intelligence with format-preserving capabilities, research and application development from 2016 to the present have adopted three main technological approaches with varying levels of complexity and effectiveness,reflecting the rapid development of the fields of natural language processing and computer vision (Hassan et al., 2020; Smith, 2019).An analysis of the research literature reveals a clear shift from traditional statistical methods to solutions based on neural networks and large language models, alongside the development of structural text analysis techniques that enable the automatic and more accurate processing of complex formatting elements (Vaswani et al., 2017; Park et al., 2021).The fundamental difference from pure text translation systems is that the task of translating formatted documents requires simultaneously addressing two distinct yet closely interrelated issues: the quality of the translation of the content and the integrity of the presentation structure,where optimizing one aspect often impacts the other (Sato, 2025; Hassan et al., 2020).

Published research on AI document translation systems has been classified into three main groups based on core technology. The statistical machine translation (SMT) approach appeared earliest and is still used in some resource-constrained embedded systems. The neural machine translation (NMT) approach, based on an encoder-decoder architecture with an attention mechanism, became the standard from 2017 after the paper "Attention Is All You Need" by Vaswani et al. The large language model-based approach has been gaining dominance since 2020 when GPT-3 demonstrated few-shot translation capability without the need for fine-tuning on specialized bilingual data (Vaswani et al., 2017; Sato, 2025).

The Statistical Machine Translation (SMT) approach is a method used before the deep learning era began, based on probabilistic models derived from large-scale parallel corpora to find the most probable translation for a given source sentence (Smith, 2019). Under this approach, the system is designed to operate by segmenting the source sentence into smaller translation units called phrases, searching for corresponding phrases in a phrase table built from training bilingual data, and combining them into a complete translation according to a target language model (Hassan et al., 2020). The main advantages of this method, highlighted by research, include transparency in the translation process, allowing for manual inspection and intervention, low computational cost during inference, suitable for resource-constrained environments, the ability to directly integrate expert-built dictionaries and linguistic rules, and no need for dedicated GPUs for deployment. However, studies also point out significant limitations, including translation quality often being inferior to NMT in most language pairs due to the inability to model long-range context, a lack of capability to handle complex grammatical structures with significant differences between source and target languages, and notably, no inherent mechanism to process formatting information accompanying the text (Park et al., 2021). Research by Hassan et al. (2020) showed that SMT systems achieved an average BLEU score 8 to 12 points lower than NMT on standard WMT test sets, while the formatting preservation rate only reached 62% in experiments with DOCX documents due to the lack of a mapping mechanism between translation units and corresponding formatting components.

6 Evaluation Criteria

Existing studies on AI-powered document translation systems show a trend toward developing multidimensional evaluation frameworks that combine quantitative and qualitative methods to simultaneously measure both the quality of the translation and the degree to which the original document's formatting is preserved, two inseparable aspects in the overall evaluation of system performance (Vaswani et al., 2017; Hassan et al., 2020). A review of the literature shows that evaluation frameworks are typically built upon five main groups of metrics, including semantic translation quality, formatting preservation, user experience, technical performance, and ethical compliance, including a combination of widely used general metrics in machine translation system evaluation such as BLEU Score, TER, and METEOR, along with domain-specific metrics reflecting the challenge of preserving document formatting, such as Format Preservation Rate, Layout Accuracy, and Structure Integrity (Smith, 2019; Park et al., 2021). The set of translation quality metrics is used in studies

used to assess the equivalence of the ideas between the machine translation and the reference translation, thereby highlighting the core challenge of balancing fidelity to the source text with naturalness in the target language, which is particularly important for technical documents that require absolute terminological accuracy (Hassan et al., 2020; Sato, 2025).

7 Conclusion

This study conducted a comprehensive review of text translation systems that preserve formatting using artificial intelligence through a systematic analysis of 40 research documents from 2016 to 2025, including both academic studies and technical reports in Vietnamese and English. The research methodology involved systematically classifying and comparing commercial products, open-source frameworks, datasets, technical approaches, and evaluation methods proposed in previous studies. The analysis results reveal three major technological trends in this field, including machine translation with the advantages of transparency and low cost but limited quality and a lack of mechanisms to preserve natural language structure, neural machine translation, which offers superior quality and supports markup-aware translation but has not yet reached the 95% format preservation threshold required by businesses, and large language models are gaining the upper hand with successful products like DeepL, which has over one million business users, Google Cloud Translation with hundreds of millions of API calls daily, and Adobe Acrobat targeting the high-end professional segment.

Regarding the product and technology deployment market, analysis reveals a clear segmentation among different user groups, with DeepL positioning itself in the high-quality segment thanks to a two-stage processing pipeline that separates content from formatting and supports 31 languages, Google Cloud Translation focuses on scalability and ecosystem integration with 135 languages, while Adobe Acrobat targets professional users with complex PDF processing needs through deep integration into the Creative Cloud ecosystem. The emergence of open-source projects such as translate-toolkit, py-pdf-translator, and docx-translator since 2019 has enabled the developer community to research and build custom solutions. Analysis of technical approaches shows that the LLM-based pipeline with temperature 0.2 and top_p 0.9 achieves the best translation quality while maintaining the highest format preservation rate in existing experiments.

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